# AI Smart Attendance System Using Facial Recognition

*Project Proposal — Web Technologies*

## 1. Overview

We want to build a web app that takes student attendance automatically using face recognition. A webcam looks at the class, recognizes each student's face, and marks them present  no manual work needed. Lecturers can also manage students, classes, and attendance records through a simple website.

This project will save time, cut down on mistakes, and make attendance more reliable.

## 2. Challenges We Want To Solve

In our university right now, attendance is taken by hand  students sign a paper sheet. After class, the lecturer counts the signatures and compares that number with a headcount of students actually present. Sometimes one student signs for a friend who isn't there, so the numbers don't match, and it's hard to find out who really signed for whom.

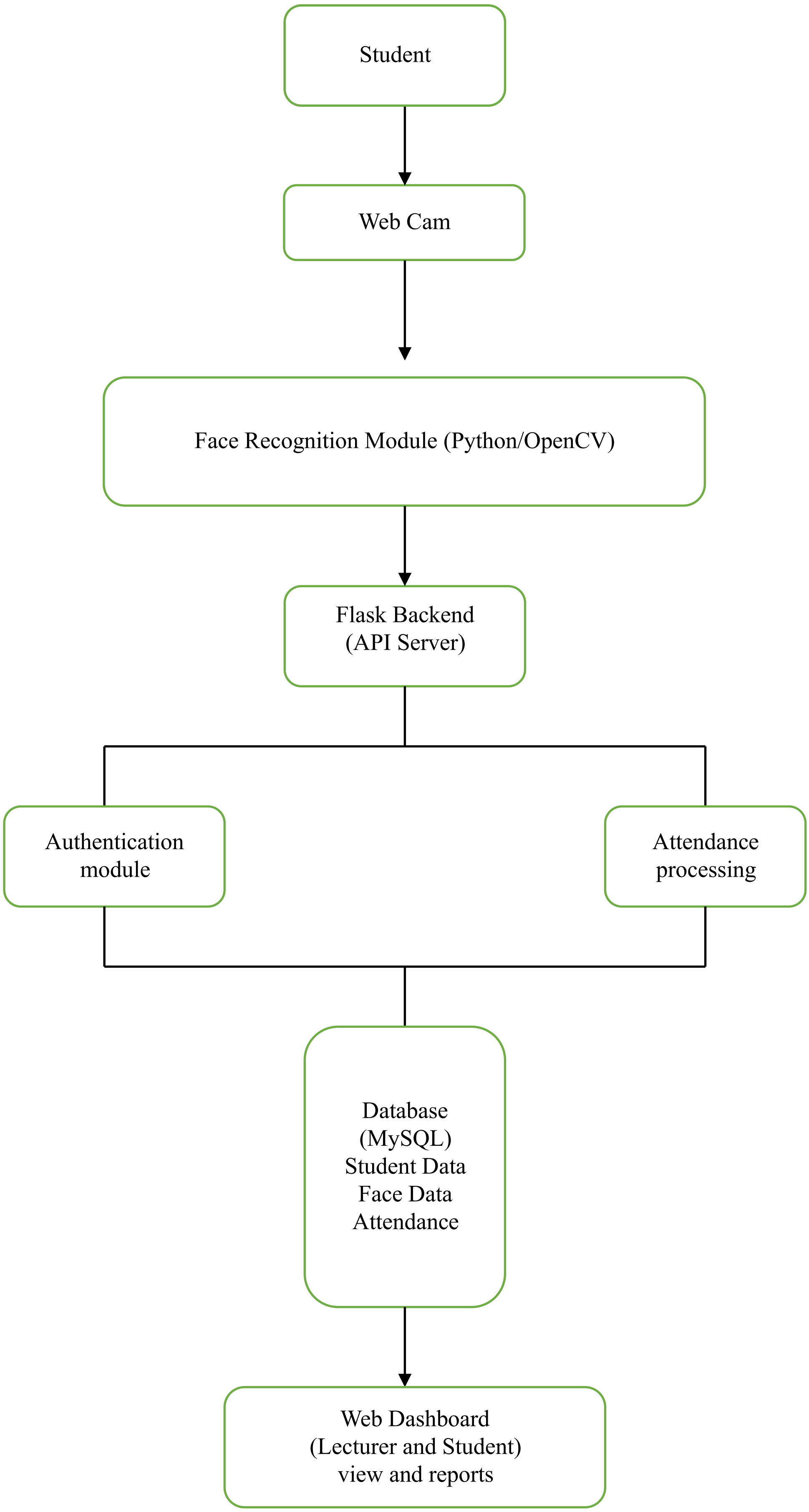
**This causes real problems:**

- Taking attendance by hand takes time out of class.
- A student can sign for an absent friend (proxy attendance).
- When the sign-count and headcount don't match, no one can easily tell who did it.
- Managing this by hand gets harder as class sizes grow.

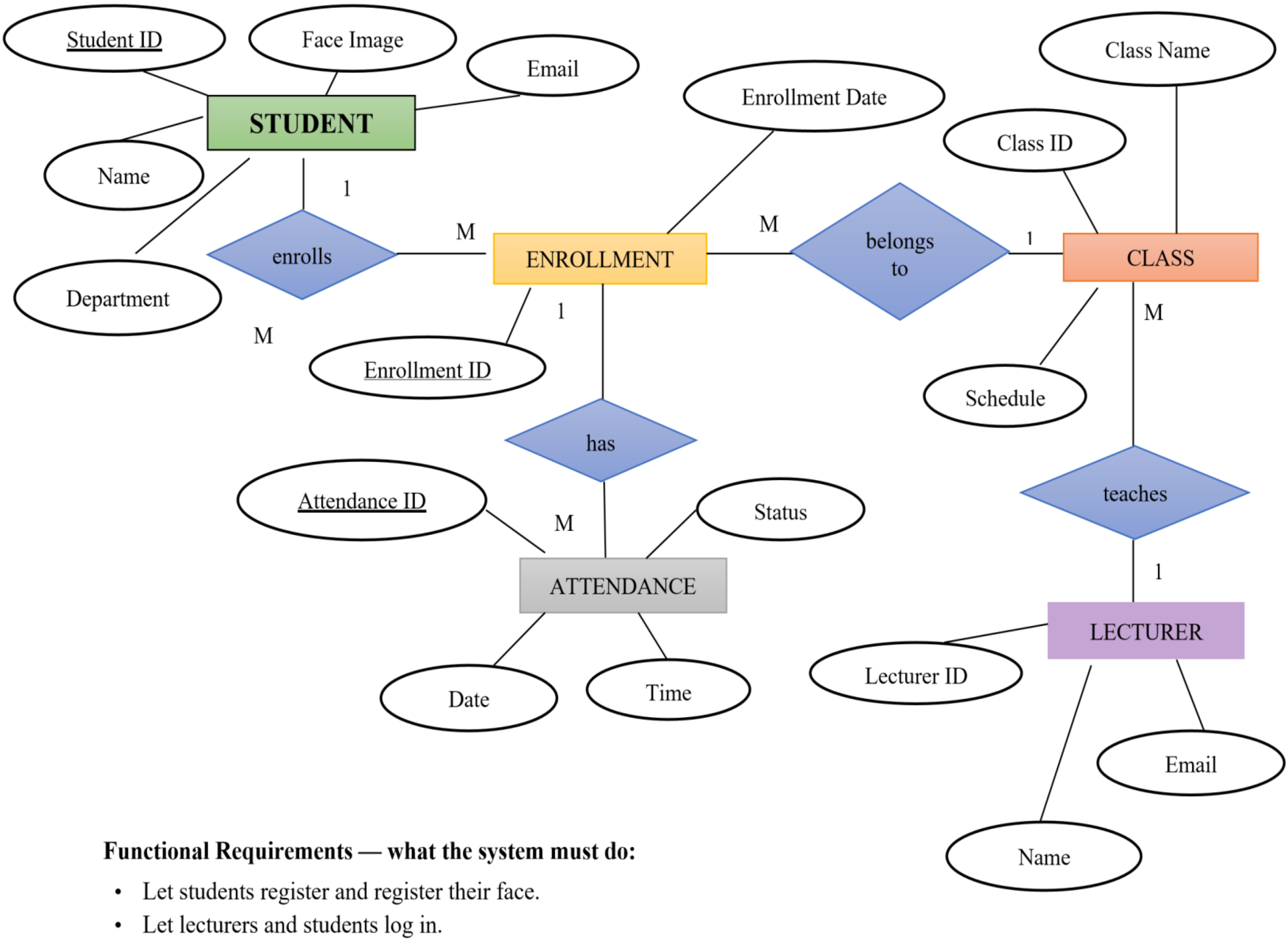
**How face recognition helps:**

- Attendance is marked automatically — no sign sheet needed.
- A student can't mark another student present, because the system checks the actual face.
- Every entry is accurate and saved digitally, so counts always match.
- Lecturers save time and get instant, trustworthy records.

## 3. System Architecture Diagram



# 4. ER Diagram




# 5. Functional and Non-Functional Requirements

**Functional Requirements – what the system must do:**

- Let students register and register their face.
- Let lecturers and students log in.

- Mark attendance automatically using face recognition.
- Let users view attendance records.
- Generate attendance reports.

**Non-Functional Requirements how the system should behave:**

- Easy to use, even for someone who isn't tech-savvy.
- Fast attendance should mark within a few seconds.
- Secure login for every user.
- Accurate and reliable attendance records.

### 6. Technology We're Using

| Part | Technology |
|---|---|
| Frontend (what students/lecturers see) | HTML, CSS, JavaScript |
| Backend (the logic behind the scenes) | Python (Flask) |
| Database (where data is stored) | MySQL or MongoDB |
| AI (the face recognition part) | OpenCV + a pre-trained face recognition library |
| Version Control (so the team can work together) | Git & GitHub |

### 7. Conclusion

The AI Smart Attendance System solves a real problem we see in our own classes slow, error-prone, manual attendance that can be gamed by proxy signing. By using face recognition, we make attendance faster, more accurate, and much harder to cheat. This project brings together web development and AI in a way that's genuinely useful for our university.

## 8. GitHub Link

https://github.com/ChamoyaVitharana245/ai-smart-attendance-system-using-facial-recognition.git